

# Pablo Garcia Fernandez

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in Pablo García Fernández

## Profile

PhD in Computer Vision from the University of Santiago de Compostela (USC) and researcher at the Centro Singular de Investigación en Tecnoloxías Intelixentes (CiTIUS). My work explores the frontiers of visual recognition under **minimal supervision**, focusing on object detection beyond closed-set assumptions. This includes open-vocabulary models, few-shot generalization, and category-free inference.

## Education

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| <b>PhD</b> | Ph.D. in <b>INFORMATION TECHNOLOGY RESEARCH</b> . <i>University of Santiago de Compostela</i> , Santiago de Compostela. <ul style="list-style-type: none"><li>• Advisor: Manuel Mucientes Molina and Daniel Cores.</li><li>• With honors, <b>cum laude</b>.</li><li>• Topics: Object detection, Vision-language models, Few-shot learning, Open-vocabulary.</li></ul> | 2022 – 2025 |
| <b>MS</b>  | M.Sc. in <b>COMPUTER VISION</b> . <i>University of Santiago de Compostela, Vigo, Coruña and Porto</i> , Santiago de Compostela. <ul style="list-style-type: none"><li>• 9.5/10   <b>Master Thesis</b>: Few-shot object detection through pseudo-label mining.</li></ul>                                                                                               | 2021 – 2022 |
| <b>BS</b>  | B.Sc. in <b>COMPUTER SCIENCE</b> . <i>University of Santiago de Compostela</i> , Santiago de Compostela. <ul style="list-style-type: none"><li>• 9.2/10   <b>Coursework</b>: Machine Learning, Computer Vision, Artificial Intelligence, HPC   <b>Bachelor Thesis</b>: Object detection and Few-shot learning.</li></ul>                                              | 2017 – 2021 |

## Research Experience

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| <b>University of Santiago de Compostela</b> , Research Fellow <ul style="list-style-type: none"><li>• Object Detection with Minimal Supervision: From Few-Shot to Open-Vocabulary.</li><li>• At Centro Singular de Investigacion en Tecnoloxias Intelixentes (CiTIUS), supervised by Manuel Mucientes.</li></ul> | Santiago de Compostela, Spain<br>2020 – Present |
| <b>University of Trento</b> , Visiting Ph.D. student <ul style="list-style-type: none"><li>• Open-vocabulary object detection on X-ray images.</li><li>• At Multimedia and Human Understanding Group (MHUG), supervised by Elisa Ricci.</li></ul>                                                                | Trento, Italy<br>2024 – 2025                    |
| <b>Gradient</b> , Intern <ul style="list-style-type: none"><li>• Deep Learning-based visual spoofing detection.</li></ul>                                                                                                                                                                                        | Vigo, Spain<br>2020 – 2020                      |

## Publications

- [1] Pablo Garcia-Fernandez, Lorenzo Vaquero, Mingxuan Liu, Feng Xue, Daniel Cores, Nicu Sebe, Manuel Mucientes, and Elisa Ricci. *Superpowering Open-Vocabulary Object Detectors for X-ray Vision*. In IEEE International Conference on Computer Vision (ICCV), 2025.
- [2] Pablo Garcia-Fernandez, Daniel Cores, and Manuel Mucientes. *Enhancing few-shot object detection through pseudo-label mining*. Image and Vision Computing 154 (2025): 105379.
- [3] Pablo Garcia-Fernandez, Daniel Cores, and Manuel Mucientes. *Exploring Open-Vocabulary Models for Category-Free Detection*. In International Conference on Computer Analysis of Images and Patterns (CAIP), 2025.

## Projects

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### **REOPEN: Leveraging Artificial Intelligence for Robust Predictive Monitoring in Process Mining (2024-2027)**

- The REOPEN project aims to improve predictive models for complex process mining by using generative models to address data issues like imbalances and anomalies. It also develops novel object detection and tracking models for video to extract activities from unstructured processes, validating them in domains like forest fires and logistics. Role: member.

### **AZOR: Search, Location and Rescue of People on Land, and Fire Fighting and Prevention (2023-2026)**

- The project addresses the development of an object detection system with RGB and thermal camera to be run on an FPGA on board a UAV for two use cases: (i) search, location and rescue of people on land; (ii) fire prevention and firefighting. Role: member.

### **Responsible AI for Process Mining 2.0 (RAI4P) (2021-2025)**

- RAI4P advances Process Mining 2.0 by predicting process features and extracting events from videos of non-digitized processes, tackling noisy and incomplete data, while following Responsible AI principles and using Natural Language Generation for explanatory analytics. Role: member.

## Scholarships and Awards

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**FPU Predoctoral Fellowship**, Spanish Ministry of Universities. Highly competitive national fellowship awarded to outstanding predoctoral candidates in Spain.

**Extraordinary Master's Award**, University of Santiago de Compostela. Recognition for top-performing students in the Master's program.

**Best Master's Thesis Award**, University of Santiago de Compostela. Awarded for the best Master's thesis in the university.

**Best Bachelor's Thesis Award**, University of Santiago de Compostela. Awarded for the best Bachelor's thesis in the university.

## Teaching

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**Automata Theory and Formal Languages**, Teaching Assistant (FPU Fellowship), University of Santiago de Compostela.

**Object-Oriented Programming**, Teaching Assistant (FPU Fellowship), University of Santiago de Compostela.